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# PUBLIC SUBMISSION

**Docket:** NSF_FRDOC_0001
Recently Posted NSF Rules and Notices.

**Comment On:** NSF_FRDOC_0001-3479
Request for Information: Development of an Artificial Intelligence Action Plan

**Document:** NSF_FRDOC_0001-DRAFT-3111
Comment on FR Doc # 2025-02305

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## Submitter Information

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## General Comment

See attached file(s)

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## Attachments

comment

Fed_governement_comment

# The necessity of foundational machine learning theory in understanding AI systems and mitigating AI risk

Mitigating AI risk has become a topic of intense effort in recent years and months. The purpose of this comment is to make a case that developing a fundamental theory of deep and other foundational model learning is a prerequisite for managing risk as our society transitions to wide use of AI technology. Theory in this context refers to identifying precise measurable quantities and mathematically describing their patterns, the way it is used in physics and engineering, rather than necessarily proving rigorous theorems. While, admittedly, even a comprehensive theory of deep learning cannot guarantee a successful AI transition, without such theory, however, we certainly would not be able to control or defend against misuse of AI systems as their behavior is already of similar or exceeding complexity compared biologicas systems or humans.

Never before had a technology been deployed so widely and so quickly with so little understanding of its fundamentals. Given the societal impact of rapidly developing AI, this is a matter of urgent importance.

In the past, mathematical theory has been "unreasonably effective" in describing and helping to engineer physical processes used in modern technology. Every electric device uses laws of electromagnetism, GPS systems use both special and general relativity to compute accurate locations, chip production and imaging technologies use quantum mechanics, the list goes on. While large amounts of hard empirical engineering work are required to design a bullet train, a space rocket, or a computer chip, we have a solid theoretical grasp of the underlying principles. Even though our understanding of AI is far less developed, there is real hope that such fundamental principles may exist and can be discovered and used to understand, monitor and control advanced AI systems. This is evidenced by recent progress in the theory of deep learning, including rethinking of overfitting and model complexity, theory of infinitely wide neural networks, optimization for over-parameterized systems and the emerging understanding of low-dimensional feature learning.

While investments in LLMs and other deep learning systems, have recently skyrocketed and now represent a major portion of all technological investments, relatively little funding goes toward basic theory of machine learning. In the past, fundamental science had often operated on a different timescale from technological development. We do not have that luxury now. Given an unprecedented pace of technological and societal change, it is hard to see what is coming in the next few months and years, let alone decades. Understanding the fundamental theory of deep learning as well as large foundational models and applying these analyses to real systems is a compelling and urgent need that can only be addressed if research funding agencies in the nation align on this key goal.

# The necessity of foundational machine learning theory in understanding AI systems and mitigating AI risk

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